Note that by (3) and (4) we see that

$$f(x+\Delta x) \approx f(x)+f'(x)\Delta x,$$

which is useful in approximating the value of $f(x+\Delta x)$.

Note that linear approximation for f at x_0 gives a good approximation to $f(x)$ if x is close to x_0, because

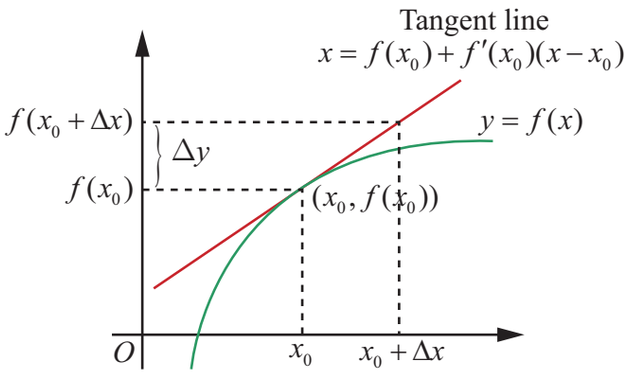


Fig. 8.3
Linear Approximation by Tangent Line

$$\text{Error} = f(x)-L(x) = f(x)-f(x_0)-f'(x_0)(x-x_0) \qquad \ldots (5)$$

approaches zero as x approaches to x_0 by continuity of f at x_0. Also, if $f(x)=mx+c$, then its linear approximation is $L(x)=(mx_0+c)+m(x-x_0)=mx+c=f(x)$, for any point $x\in(a,b)$. That is, the linear approximation, in this case, is the original function itself (is it not surprising?).

Example 8.1

Find the linear approximation for $f(x)=\sqrt{1+x}, x\geq -1$, at $x_0=3$. Use the linear approximation to estimate $f(3.2)$.

Solution

We know from (4), that $L(x)=f(x_0)+f'(x_0)(x-x_0)$. We have $x_0=3, \Delta x=0.2$ and hence $f(3)=\sqrt{1+3}=2$. Also,

$$f'(x) = \frac{1}{2\sqrt{1+x}} \text{ and hence } f'(3)=\frac{1}{2\sqrt{1+3}}=\frac{1}{4}.$$

$$\text{Thus, } L(x) = 2+\frac{1}{4}(x-3)=\frac{x}{4}+\frac{5}{4} \text{ gives the required linear approximation.}$$

$$\text{Now, } f(3.2) = \sqrt{4.2} \approx L(3.2)=\frac{3.2}{4}+\frac{5}{4}=2.050.$$

Actually, if we use a calculator to calculate we get $\sqrt{4.2}=2.04939$. ■

8.2.2 Errors: Absolute Error, Relative Error, and Percentage Error

When we are approximating a value, there occurs an error. In this section, we consider the error, which occurs by linear approximation, given by (4). We shall consider different types of errors. Taking $h=x-x_0$, we get $x=x_0+h$, then (5) becomes

$$E(h) = f(x_0+h)-f(x_0)-f'(x_0)h. \qquad \ldots (6)$$

Note that $E(0)=0$ and as we have already observed $\lim_{h\to 0} E(h)=0$ follows from the continuity of f at x_0. In addition, if f is differentiable, then from (1), it follows that

Chapter 8 Differentials and Partial Derivatives.indd 61 26-02-2020 17:00:57